Interpretable ESPI Mode Classification under Data Scarcity with Frequency-Prior Hybrid Random Forests

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Abstract

We study the automated classification of vibration modes in bouzouki-family instruments from time-averaged Electronic Speckle Pattern Interferometry (ESPI) images under severe class imbalance, limited data availability, and cross-board domain shift. Instead of relying exclusively on end-to-end convolutional neural networks, we propose a frequency-prior Hybrid Random Forest operating on a 22-dimensional feature space that combines 16 topology-oriented pattern descriptors with 6 physically motivated frequency priors. On a curated dataset of 3,443 ESPI images, the proposed model achieves 97.85% accuracy and 95.15% Macro-F1, outperforming both a pattern-only Random Forest (90.15% / 69.91%) and a ResNet-18 baseline (93.76% / 88.11%). Feature-importance analysis shows that 96.86% of the Gini-based discriminative power is attributed to the frequency priors; held-out permutation importance yields a more conservative but consistent estimate (≈82.5% frequency / ≈17.5% morphology), providing empirical support for a functional disentanglement hypothesis that separates material-invariant topological information from material-dependent spectral characteristics. Additional Leave-One-Band-Out and Leave-One-Dataset-Out experiments indicate that the proposed representation improves robustness to unseen frequency bands, while strong board-level domain shift remains an open challenge. These findings suggest that frequency-prior feature engineering offers a competitive and interpretable alternative to black-box CNNs for ESPI-based modal classification in low-data regimes.

CCS Concepts

• **Computing methodologies** → **Machine learning approaches**; *Image processing*; *Neural networks*; • **Information systems** → Data mining.

Keywords

frequency priors, ESPI, vibration mode classification, interpretability, feature engineering, random forest

1 Introduction

Electronic Speckle Pattern Interferometry (ESPI) is a full-field optical technique for capturing vibration mode shapes through fringe patterns, and time-averaged ESPI is particularly suitable for modal analysis of instrument soundboards [1, 2]. In bouzouki-family instruments, the identification of modal patterns such as the Helmholtz mode, the fundamental transverse mode, and higher-order responses is relevant to structural and vibroacoustic assessment.

Automating this task with machine learning is nontrivial because ESPI classification combines several difficulties that rarely appear together in standard vision benchmarks. First, the available dataset is relatively small, with 3,443 labelled images after quality control and curation. Second, the class distribution is highly imbalanced: the minority class mode 2,1 contains 58 samples (1.7%), whereas the majority class other/unknown contains 1,794 samples (52.1%), corresponding to an imbalance ratio of 30.91:1. Third, board-specific variability introduces substantial domain shift, especially for specimen W02.

A straightforward end-to-end deep-learning approach is therefore not necessarily optimal. In this study, a ResNet-18 baseline performs strongly, yet remains inferior to a frequency-prior alternative in both accuracy and Macro-F1. This motivates a different formulation of the problem. Rather than treating ESPI modal identification as generic image classification, we model it as a constrained physical inference problem in which topological information from fringe maps is combined with physically motivated frequency priors informed by thin-plate vibration theory [3, 4].

To this end, we propose a Hybrid Random Forest classifier that fuses morphology-based pattern features with explicit frequency priors. The resulting model reaches 97.85% accuracy and 95.15% Macro-F1, compared with 90.15% and 69.91% for a pattern-only Random Forest and 93.76% and 88.11% for a ResNet-18 baseline. Feature-importance analysis further shows that the six frequency priors dominate the decision process, accounting for 96.86% of total Gini importance, while held-out permutation importance confirms this hierarchy at a more conservative level (≈82.5%).

The contributions of this paper are fourfold:

(1) We introduce a frequency-prior Hybrid Random Forest for ESPI mode classification that combines 16 handcrafted pattern features with 6 frequency priors.
(2) We provide a structured comparison from a frequency-metadata baseline to a pattern-only Random Forest and the full Hybrid Random Forest, thereby isolating the contribution of morphology and frequency priors in the main experimental branches.
(3) We compare the proposed model against a strong ResNet-18 baseline and observe consistent gains in both overall accuracy and Macro-F1.
(4) We evaluate robustness with Leave-One-Band-Out (LOBO) and Leave-One-Dataset-Out (LODO) protocols, thereby distinguishing frequency-wise robustness from cross-board domain generalization.

2 Related Work

Prior work on ESPI and modal analysis has traditionally followed two broad directions: handcrafted descriptors of fringe geometry and direct image-based classification with convolutional architectures. In small and noisy datasets, however, purely data-driven approaches are vulnerable to overfitting and to shortcut learning based on nuisance correlations rather than physically meaningful structure. ESPI images pose particular challenges for standard vision methods: they exhibit multiplicative speckle noise, low global

contrast, and fringe patterns whose spatial frequency varies continuously with vibration amplitude, making texture-based features unreliable without domain-specific preprocessing.

More broadly, physics-informed machine learning has shown that incorporating domain knowledge can improve both generalization and interpretability, particularly when the available training data are limited [7]. The dominant paradigm in this area embeds physics through PDE-constrained loss functions or neural operator architectures. In the present setting, however, the relevant form of physics guidance is simpler and arguably more natural: explicit feature design grounded in vibration theory and modal frequency structure. This places the approach closer to the tradition of physics-informed feature engineering in fluid mechanics and materials science, where theory-derived descriptors have been shown to outperform purely data-driven alternatives in low-data regimes.

In mode-shape recognition, Zernike moment descriptors have been used for vibration pattern classification in mechanical structures [10], and more recent work has extended Zernike annular moments with Random Forest and gradient-boosted classifiers for tire modal analysis [11]. For musical instrument acoustics specifically, Brezas et al. [12] combined ESPI measurements with finite-element modeling and psychoacoustic evaluation for carbon fiber bouzouki soundboard assessment, though without automated classification. To the best of our knowledge, no prior work has attempted automated machine learning classification of ESPI fringe patterns by vibration mode type. The present study addresses this gap.

The representation-learning literature also motivates a disentangled view of the problem. Here, however, the notion is used in a functional rather than a latent-variable sense. The proposed feature space separates topology-oriented pattern information from spectral information carried by the excitation frequency and by mode-specific distance-to-interval priors. The analysis therefore does not claim formal disentanglement in the generative-modeling sense. Instead, it offers empirical support for a functional separation between material-invariant topological structure and material-dependent spectral cues.

3 Dataset and Problem Formulation

The classification dataset consists of 3,443 curated ESPI images obtained after preprocessing, quality control, and expert label verification. The six classes are mode 1,1H, mode 1,1T, mode 1,2, mode 2,1, mode higher, and other/unknown. Their supports are 199, 171, 106, 58, 1,115, and 1,794 samples, respectively. The modal labels follow a (m, n) indexing convention, where m and n denote the number of nodal lines along the longitudinal and transverse axes of the soundboard. Mode 1,1H and mode 1,1T correspond to the first transverse mode coupled with Helmholtz cavity resonance and the first purely transverse mode, respectively; mode 1,2 and mode 2,1 introduce additional nodal complexity; and mode higher aggregates all modes above the fundamental group. The other/unknown class contains images where no clear modal pattern was identified by expert annotators, including images acquired at frequencies that do not excite a clean resonance. Figure 1 provides representative time-averaged ESPI fringe maps for the six classes. These examples are included to ground the classification problem visually and to illustrate why explicit topology-oriented descriptors remain useful alongside the frequency priors.

Table 1: Class distribution in the curated ESPI dataset.

Class	Samples	Share (%)
Mode 1,1H	199	5.78
Mode 1,1T	171	4.97
Mode 1,2	106	3.08
Mode 2,1	58	1.68
Mode higher	1,115	32.38
Other/unknown	1,794	52.11

The dataset is also unevenly distributed across boards. W01 contributes 502 samples, W02 contributes 2,803 samples, and W03 contributes 138 samples. This matters because board identity is entangled with material properties, acquisition conditions, and speckle behavior, making cross-board generalization considerably harder than in-distribution classification.

We formulate the task as supervised multiclass classification. Each sample consists of a fringe map together with associated frequency metadata, and the goal is to predict the modal class label. The main in-distribution evaluation uses a 70/15/15 train/validation/test split. This evaluation is complemented by bootstrap confidence intervals as well as by LOBO and LODO protocols.

This distribution has two direct implications for model design. First, overall accuracy alone is insufficient, because a classifier biased toward the majority class can still appear competitive. Second, minority-class reliability becomes a central requirement, especially for physically meaningful but sparsely represented modes. These considerations motivate both the choice of Macro-F1 as a primary metric and the emphasis on interpretable models whose failure modes can be inspected.

4 Methodology

4.1 Feature Representation

The proposed Hybrid Random Forest operates on a 22-dimensional feature vector. This vector combines 16 pattern features extracted from ESPI fringe maps with 6 frequency-based priors.

The pattern features summarize spatial and morphological structure in the fringe map. They include descriptors related to fringe geometry, symmetry, nodal topology, region-of-interest statistics, and FFT-based spectral characteristics. Representative examples include total fringe count, average fringe spacing, centroid coordinates, valid-pixel ratio, area extent, nodal length, nodal complexity, Euler number, symmetry score, FFT peak magnitude, energy concentration, and spectral entropy. The 16 pattern features can be grouped into five interpretable families. Fringe-density descriptors quantify the richness and spacing of the interference structure through quantities such as total fringe count, estimated counts along the principal axes, and average fringe spacing. Geometry- and symmetry-related descriptors capture the global organization of the pattern through centroid location, horizontal-to-vertical and diagonal ratios, and an aggregate symmetry score. Moment-based descriptors summarize the overall shape of the active fringe regions

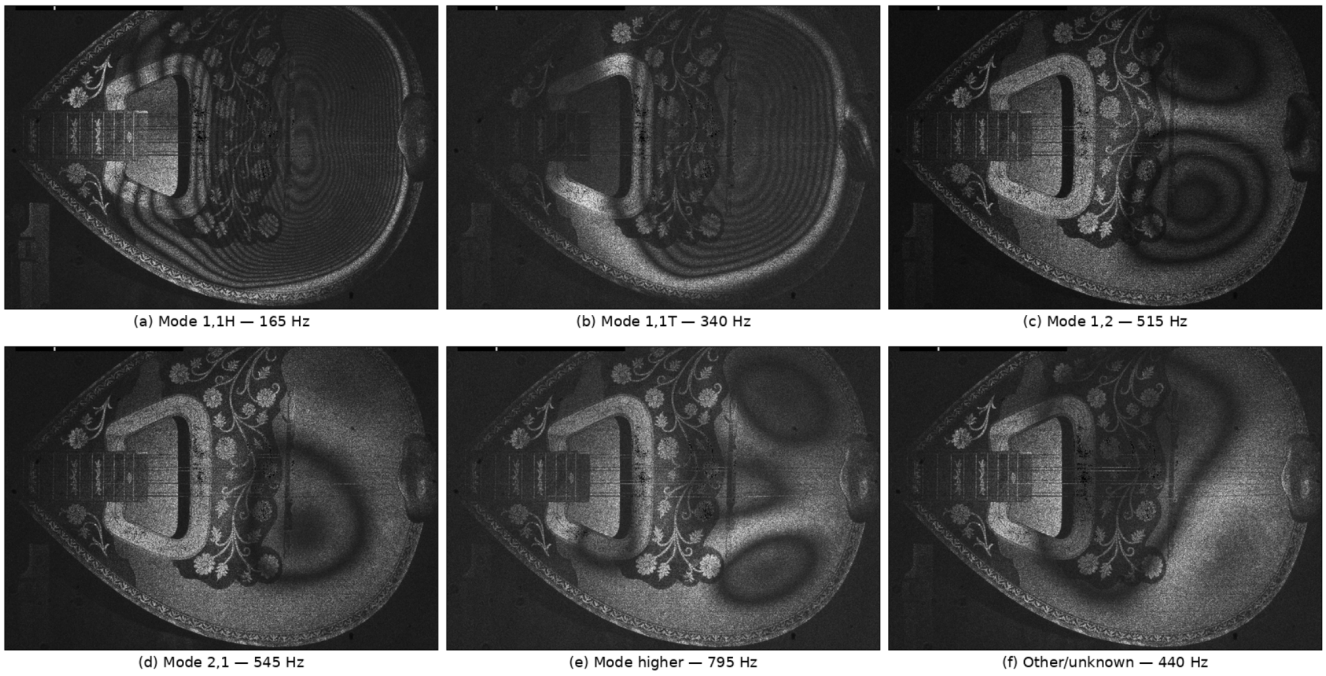


Figure 1: Representative time-averaged ESPI fringe maps for the six vibration mode classes used in this study. Modes 1,1H, 1,1T, 1,2, and 2,1 follow the modal indexing convention adopted in the paper; the higher class aggregates more complex patterns above the fundamental modal group, whereas the other/unknown class contains ambiguous or non-canonical fringe patterns for which no clean modal assignment was made. The broader laser-based interferometric framework from which such examples originate is described by Bakarezos et al. [13].

in a translation- and scale-aware manner. Nodal-topology descriptors characterize structural complexity through nodal length, Euler number, and nodal complexity. Finally, FFT-based spectral descriptors encode the distribution of spatial-frequency energy, including energy concentration and spectral entropy. This grouping reflects a decomposition that parallels, in broad terms, the interpretive cues applied during manual modal inspection.

The six frequency priors encode information derived from the excitation frequency and from expected modal frequency structure. These include the excitation frequency itself together with mode-specific distance features to the theoretical frequency intervals associated with the dominant modal classes. The intervals used in the underlying experimental study are 155–175 Hz for mode 1,1H, 320–345 Hz for mode 1,1T, 500–525 Hz for mode 1,2, 540–570 Hz for mode 2,1, and 680–1500 Hz for mode higher.

This design reflects a functional separation between topology and spectrum. Pattern features describe the structure of the observed mode shape, while frequency priors constrain the set of plausible classes in a physically meaningful way.

More concretely, each prior can be interpreted as a soft distance-to-interval penalty. If f is the excitation frequency and $I_k = [a_k, b_k]$ is the expected interval for class k, the corresponding prior can be written as

$$d_k(f) = \begin{cases} 0, & f \in I_k, \\ \min\{|f - a_k|,\ |f - b_k|\}, & f \notin I_k. \end{cases} \tag{1}$$

This formulation is useful because it does not force hard thresholding. Instead, it lets the classifier combine spectral plausibility with morphology. In transition regions where neighboring modes overlap, the pattern features can still contribute to the final decision.

4.2 Classification Models

Three classification branches are central to the present comparison. The first is a frequency-metadata baseline using only the excitation frequency and its normalized form. The second is a pattern-only Random Forest using the 16 handcrafted pattern features. The third is the full Hybrid Random Forest combining the pattern features with the six frequency priors. In addition, the originating thesis reports a supplementary frequency-only ablation based on the six physics priors; this result is discussed qualitatively but is not treated as one of the main experimental branches in the present paper.

The Random Forest implementation uses 100 estimators, balanced class weighting, `max_features` = $\sqrt{d}$, `min_samples_leaf` = 2, and `random_state` = 42. This setup is selected to address severe class imbalance while preserving interpretability through impurity-based feature ranking [5].

As a deep-learning baseline, the study also evaluates a ResNet-18 classifier trained on raw pixel inputs [6]. More specifically, the baseline uses an ImageNet-pretrained ResNet-18 whose input layer is adapted to single-channel grayscale images and whose final fully connected layer is replaced to produce six output classes. In the

thesis implementation, Layers 1–2 were frozen while the higher layers were optimized for the target task, a choice motivated by the small-data regime and the observed tendency of deeper networks to overfit. Training used mini-batches of 64 images. We do not claim that this baseline exhausts the full CNN design space; rather, it serves as a strong and practically reasonable data-driven reference against which the proposed frequency-prior representation can be evaluated. In this dataset regime, ResNet-18 is preferred over deeper variants because deeper models such as ResNet-50 and ResNet-101 showed stronger overfitting tendencies.

4.3 Evaluation Protocols

The primary in-distribution evaluation uses a 70/15/15 split. Accuracy, Macro-F1, Weighted-F1, and per-class precision, recall, and F1-score are reported. Macro-F1 is particularly important here because of the 30.91:1 imbalance ratio.

To quantify uncertainty, we additionally report bootstrap confidence intervals based on 1,000 resamples. For robustness beyond the standard split, two group-based protocols are used. In LOBO, the data are partitioned into 111 frequency bands and each held-out band is tested against models trained on the remaining bands. In LODO, each board is held out in turn and testing is performed on the unseen board. The statistical analysis also includes paired comparisons between the Hybrid Random Forest and its baselines. In the underlying study, McNemar testing indicates significant superiority of the Hybrid Random Forest over the pattern-only baseline ($p < 0.001$), with a very large accompanying effect size reported in the originating thesis ($d = 2.89$), while comparison against ResNet-18 also indicates significant superiority.

In the originating thesis, an upstream denoising stage was also examined; however, the present paper focuses on classification rather than on the full denoising–classification chain. That broader analysis showed that denoising quality and downstream classification quality do not always align: improvements in image-level criteria such as PSNR and SSIM do not automatically translate into better modal classification. This task-misalignment observation is important because it justifies evaluating preprocessing functionally, through the quality of the final modal decision, rather than exclusively through generic image-restoration metrics.

5 Results

5.1 Main In-Distribution Comparison

Table 2 summarizes the main classification results. The frequency-metadata baseline reaches 55.30% accuracy and 22.00% Macro-F1. The pattern-only Random Forest improves substantially to 90.15% accuracy and 69.91% Macro-F1. The ResNet-18 baseline achieves 93.76% accuracy and 88.11% Macro-F1. The best model is the Hybrid Random Forest, which reaches 97.85% accuracy, 95.15% Macro-F1, and 97.87% Weighted-F1.

These gains are not marginal. Relative to the pattern-only Random Forest, the Hybrid Random Forest improves Macro-F1 by 25.24 percentage points. Relative to ResNet-18, it improves accuracy by 4.09 percentage points and Macro-F1 by 7.04 percentage points. The 95% bootstrap confidence interval for Hybrid Random Forest accuracy is 97.36–98.34, while the corresponding 95% confidence interval for Macro-F1 is 93.97–96.24.

Table 2: Main in-distribution comparison.

Model	Acc. (%)	M-F1 (%)	W-F1 (%)
Freq.-metadata RF	55.30	22.00	50.00
Pattern-only RF	90.15	69.91	88.53
ResNet-18	93.76	88.11	92.30
Hybrid RF	**97.85**	**95.15**	**97.87**

Table 3: Per-class performance of the Hybrid RF, with class supports from the curated dataset (N = 3,443).

Class	Prec. (%)	Rec. (%)	F1 (%)	n
Mode 1,1H	100.00	92.46	96.08	199
Mode 1,1T	88.14	100.00	93.70	171
Mode 1,2	82.41	83.96	83.18	106
Mode 2,1	100.00	100.00	100.00	58
Mode higher	100.00	100.00	100.00	1,115
Other/unk.	98.21	97.66	97.93	1,794
Macro avg	94.79	95.68	95.15	3,443

The ablation ladder is informative in its own right. The large jump from the frequency-metadata baseline to the pattern-only Random Forest shows that fringe topology contains substantial discriminative information beyond the naive use of excitation-frequency metadata alone. Additional thesis ablations further indicated that a frequency-only Random Forest captures most of the discriminative signal, while the Hybrid model remains the strongest overall performer. This pattern supports the interpretation that spectral priors provide the dominant axis of discrimination, whereas morphology contributes complementary information that becomes especially useful under class imbalance and in modal transition regions where neighboring frequency intervals overlap.

5.2 Per-Class Performance and Feature Importance

Per-class results further justify the use of Macro-F1 as the primary metric. Table 3 reports per-class precision, recall, and F1-score for the Hybrid RF, together with the corresponding class supports in the curated dataset. Particularly notable are the perfect scores for mode 2,1 despite its minority status (58 samples), and the comparatively lower F1 of mode 1,2, which occupies a transitional frequency region (500–525 Hz) adjacent to mode 2,1 (540–570 Hz).

Interpretability is one of the strongest aspects of the proposed method. Two complementary importance measures are reported to provide a robust assessment of feature relevance.

Gini importance (MDI). In the Hybrid Random Forest, the six frequency priors account for 96.86% of total Gini importance, whereas the 16 pattern features account for only 3.14%. The top-ranked features are the distance to mode 1,1H, the excitation frequency, the distance to mode 1,1T, the distance to mode 1,2, the distance to mode 2,1, and the distance to mode higher.

Table 4: Generalization results under LOBO.

Protocol	Model	Acc. (%)	M-F1 (%)
LOBO	Pattern-only RF	83.70	76.20
LOBO	ResNet-18	87.20	82.10
LOBO	Hybrid RF	91.83	88.31

Held-out permutation importance. Because Gini importance is known to be biased toward continuous features with many candidate split points [8], we additionally compute permutation importance on the held-out evaluation set. When grouped by feature category, the frequency priors account for ≈82.5% of total permutation importance and the morphology features for ≈17.5%. This more conservative estimate is consistent with the Gini-based ranking but assigns substantially more credit to the pattern descriptors.

The gap between the two measures is itself informative. The Gini estimate (96.86% frequency) reflects the well-documented tendency of MDI to concentrate importance mass on the features that are selected earliest and most frequently in the tree ensemble, particularly when correlated encodings of the same underlying signal are present [9]. The permutation estimate (≈82.5% frequency) corrects for this bias by measuring the actual predictive contribution of each feature on unseen data. Both measures converge on the same qualitative conclusion: frequency priors are the dominant axis of discrimination, while morphology provides a non-negligible complementary contribution.

This does not imply that pattern features are irrelevant; rather, it indicates that once physically meaningful spectral constraints are made available, they become the dominant axes of discrimination. A useful interpretation is that morphology carries the information needed to recognize the spatial signature of a mode, whereas the priors place that signature in a spectrally plausible context. This is precisely the kind of division of labor one would expect from a physically informed representation.

The result is also important from the standpoint of explainable AI. In a black-box CNN, errors can be difficult to diagnose because the effective decision variables remain latent. In contrast, the Hybrid Random Forest provides a direct inspection path: one can identify whether a wrong decision originated from an ambiguous fringe pattern, an outlying excitation frequency, or a mismatch between the two. For experimental disciplines such as optical metrology, that level of interpretability is not merely convenient; it is methodologically useful.

5.3 LOBO and LODO Generalization

The LOBO protocol provides a more realistic view of generalization across unseen frequency regions. Across 111 held-out frequency bins, the pattern-only Random Forest achieves 83.70% LOBO accuracy and 76.20% Macro-F1, ResNet-18 achieves 87.20% and 82.10%, and the Hybrid Random Forest achieves 91.83% and 88.31%.

The Hybrid Random Forest also has the largest number of perfect bins (62/111) and the fewest problematic bins below 60% accuracy (12/111). The main problematic regions are modal transition zones, especially around 175–180 Hz, 320–360 Hz, and 495–555 Hz. The worst reported LOBO bin is 175–180 Hz, with an accuracy of 4.8%.

Table 5: Hybrid RF performance for each held-out board under LODO.

Held-out board	Accuracy (%)	Macro-F1 (%)
W01	83.67	56.76
W02	15.27	8.21
W03	100.00	100.00

Even so, the Hybrid model degrades more gracefully overall than both baselines, supporting the claim that frequency priors improve robustness under frequency shift.

The LODO results constitute the main limitation of the method. When W01 is held out, the Hybrid Random Forest achieves 83.67% accuracy and 56.76% Macro-F1. When W02 is held out, performance drops sharply to 15.27% accuracy and 8.21% Macro-F1. When W03 is held out, the reported performance is 100.00% accuracy and 100.00% Macro-F1.

The asymmetry across held-out boards has a structural explanation. W02 contributes 81.4% of all samples; when it is held out, the remaining training set consists of only 640 images from W01 and W03, which may not span the full variability of speckle behavior and modal expression present in W02. Conversely, when W03 is held out, the model retains 96.0% of the data, and the small size of W03 (138 samples) means that its modal patterns are well covered by the training distribution. The W01 result (83.67% accuracy, 56.76% Macro-F1) represents an intermediate case where the model generalizes reasonably for the majority classes but fails on underrepresented modes. In particular, the W02 collapse suggests that the current representation, although highly effective in-distribution and under frequency shift, is not yet sufficient for robust board-level transfer when the dominant specimen is excluded from training. This observation points to two concrete directions for future work: board-aware calibration of the spectral priors and explicit domain-adaptation strategies across specimens.

Across the three held-out boards, the mean LODO accuracy is 66.31 ± 44.11, while the mean LODO Macro-F1 is 55.00 ± 46.81. These numbers show that board-level generalization is substantially harder than frequency-wise generalization, and that the high variance is driven primarily by the extreme sensitivity to W02 exclusion.

6 Discussion

Taken together, the results show that explicit frequency-prior feature engineering can be highly effective for ESPI classification under small-data and imbalanced conditions. The Hybrid Random Forest consistently outperforms both a pattern-only Random Forest and a ResNet-18 baseline on the main split, improves Macro-F1 substantially, and remains computationally lightweight compared with deep-learning alternatives.

The main conceptual takeaway is that physically meaningful priors can be more valuable than additional model complexity. In this dataset, discrimination between modal classes depends heavily on spectral structure, and the Hybrid Random Forest benefits from direct access to that structure through theory-guided frequency features. This also explains why the model is more interpretable

than the CNN baseline: its dominant decision variables can be inspected, ranked, and linked back to vibration theory.

From a representation perspective, the findings provide empirical support for a functional disentanglement hypothesis: spectral priors encode mode-specific frequency information, whereas handcrafted morphology captures the spatial structure of the fringe pattern. That formulation is intentionally more conservative than claiming formal disentanglement in the latent-variable sense. The quantitative evidence supports this: both Gini importance (96.86% frequency) and held-out permutation importance (≈82.5% frequency) indicate that frequency priors are the dominant discriminative axis, while the gap between the two measures reflects the well-known inflation of MDI for correlated continuous predictors [8]. The permutation-based estimate provides a more realistic picture of the frequency–morphology balance and confirms that pattern features retain a non-trivial role (≈17.5%), particularly in modal transition zones where frequency information alone is ambiguous.

The ablation ladder further supports this interpretation. Supplementary thesis ablations indicate that a frequency-only Random Forest captures most of the discriminative signal, confirming that spectral structure carries the majority of the predictive information. At the same time, the Hybrid model remains the strongest overall performer, which argues against a trivial frequency-lookup interpretation. The morphological component adds value precisely where it matters most: in overlapping frequency regions and in minority classes where frequency cues alone are insufficient.

At the same time, the LODO results impose an important qualification. Board-specific variability remains a serious challenge, and the W02 case shows that physically informed tabular features alone do not fully resolve provenance bias. A plausible explanation is that W02 behaves as a strong out-of-distribution case due to domain-specific characteristics, including different speckle behavior, acquisition variability, and calibration mismatch. This suggests that future work should focus on domain adaptation, board-aware calibration, or hybrid feature learning rather than assuming that stronger classifiers alone will solve the problem.

A second important observation concerns evaluation methodology. If the problem were judged only on the main train/test split, one might conclude that the classification problem has largely been solved. LOBO and especially LODO show that this conclusion would be premature. Frequency-wise robustness is strong but not perfect, and true cross-board generalization remains fragile. This distinction matters because it separates interpolation over familiar acquisition regimes from genuine transfer across new physical instances.

The originating thesis also offers a cautionary result from the preprocessing stage: denoising improvements measured with image-restoration metrics do not necessarily improve downstream recognition. That negative result is relevant here because it reinforces the broader message of the paper. In scientific imaging pipelines, model selection should be aligned with the end task. For ESPI-based modal classification, the decisive criterion is not whether an image looks cleaner in isolation, but whether the resulting representation preserves the physically meaningful cues required for reliable class separation.

A final practical implication is that the Hybrid Random Forest should not be viewed merely as a baseline to be replaced by a larger network. In this setting, its strengths—strong accuracy, transparent decision rules, modest data requirements, and straightforward error analysis—are central advantages. From an operational perspective, Random Forest training is substantially lighter than CNN retraining and is easier to integrate into iterative laboratory workflows where models may need to be updated after each acquisition campaign. A larger model may still be valuable in future work, but only if it improves transfer across boards without sacrificing these properties.

7 Limitations and Threats to Validity

Several limitations should be made explicit. First, the dataset remains modest in scale despite the fact that it is unusually rich for this application domain. The strong in-distribution performance of the proposed model should therefore not be interpreted as evidence that the problem is solved in a broad sense. Rather, it shows that physically structured representations can be highly effective when data are limited and the inductive bias is well matched to the domain.

Second, the present work focuses on a curated six-class problem formulation derived from a specific experimental workflow. This is an advantage for methodological clarity, but it also means that the observed gains are tied to the labeling protocol, the quality-control procedure, and the frequency ranges used in the study. A different acquisition protocol or a different operational taxonomy could change the relative difficulty of the classes and, in turn, alter the balance between morphology and spectral priors.

Third, the reported feature-importance values should be interpreted with care. Gini importance is useful because it offers an immediate model-level explanation, but it is not the only possible notion of relevance. Correlated predictors can redistribute importance mass, and interval-based priors can become especially dominant when the dataset is structured around physically meaningful frequency neighborhoods. The supplementary permutation-importance analysis partially addresses this concern; however, the grouped estimates (≈82.5% / ≈17.5%) should still be read as evidence about the behavior of the present model family rather than as a universal statement about the intrinsic information content of the measurements.

Fourth, the generalization analysis is informative but still incomplete. LOBO measures robustness to unseen frequency regions and LODO measures transfer to unseen boards, yet neither protocol fully captures all plausible deployment scenarios. In practice, domain shift may arise simultaneously from material variation, acquisition noise, illumination changes, calibration drift, and differences in experimental mounting conditions. A useful next step would be to design evaluation protocols that isolate these factors more systematically.

Finally, the paper is intentionally centered on classification rather than on the full denoising–classification chain. This choice keeps the contribution focused and better aligned with the strongest empirical result, namely the Hybrid Random Forest with physics priors. At the same time, it leaves open an important research question: how should preprocessing be optimized when the end goal is not visual enhancement but robust physical inference? The originating thesis suggests that this question is nontrivial, because

better image-restoration scores do not necessarily imply better modal recognition.

Beyond the specific ESPI application, the study illustrates a broader principle for scientific machine learning. When the target phenomenon is constrained by well-understood theory, carefully designed intermediate representations can outperform heavier end-to-end pipelines, especially in small-data regimes. In such settings, the most effective inductive bias is often not architectural depth but the right decomposition of the problem into physically meaningful variables.

8 Conclusion

We presented a frequency-prior Hybrid Random Forest for automated classification of vibration modes in bouzouki-family instruments from time-averaged ESPI images. The model combines 16 handcrafted pattern features with 6 physically motivated frequency priors and achieves 97.85% accuracy and 95.15% Macro-F1 on a curated dataset of 3,443 images. It outperforms both a pattern-only Random Forest and a ResNet-18 baseline, while also offering a transparent decision process through dual feature-importance analysis (Gini and permutation).

The empirical evidence further supports a functional disentanglement hypothesis in which topological and spectral information play distinct but complementary roles. LOBO results indicate improved robustness to unseen frequency bands, whereas LODO results reveal that strong cross-board domain shift remains unresolved, particularly for W02. Overall, the study suggests that frequency-prior feature engineering is a strong and interpretable direction for ESPI-based modal analysis, especially in low-data and imbalanced regimes.

A natural next step is to combine the current frequency-prior representation with explicit domain-adaptation methods targeting board-level shift. Possible directions include board-aware calibration, feature alignment across specimens, and hybrid models that preserve the interpretability of tabular priors while learning more transferable representations from fringe maps. Another promising direction is to extend the current framework to additional instrument families and broader ESPI acquisition settings, thereby testing how far the present priors generalize beyond the current dataset.

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